

From Triangles to Individual Claims: Case Estimates, Censoring, and Reserve Information

A controlled experiment in macro and micro non-life claims reserving

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Abstract. This note studies, in one controlled synthetic portfolio, a practical source of divergence between aggregate and individual claims reserving: open reported claims for which the insurer holds a case estimate but future payments remain unknown. Rolling pseudo-valuation dates create out-of-time tests against a known hidden future. Paid Chain Ladder, incurred Chain Ladder, paid-only claim-level models, raw outstanding case liabilities, and a case-aware claim-level model are compared on a common four-date window. In this experiment, the case estimate is a strong ranking signal but a poor level estimate. Naive aggregation performs badly, whereas a log-scale claim model with historically estimated Duan retransformation achieves 9.4% total-reserve MAPE, compared with 13.2% for paid Chain Ladder on this portfolio. On RBNS alone, the corresponding MAPE is 9.2%, compared with 19.3% for the paid-only micro model. These four correlated dates are descriptive evidence, not an inferential sample. The note is deliberately pedagogical: censoring, RBNS/IBNR, paid versus incurred triangles, retransformation, and evaluation metrics are developed alongside the experiment. The synthetic claims experience was generated with SynthETIC; the incurred-loss and outstanding case liability (OCL) process was supplied by SPLICE. This distinction matters because the behaviour of case estimates is itself part of the experiment. A fixed-model diagnosis of the year-10 miss shows material conditional miscalibration across OCL bands arising from both log-scale mean-function misspecification and heteroscedastic residuals. A final added-variable diagnostic shows that the case estimate contains substantial incremental information about future RBNS level beyond the paid-state variables used here: after partialling OCL on paid-to-date, claim age and report age, the residual OCL signal has pooled partial Spearman correlation 0.544 with the date-debiased paid-model error, with year-specific values 0.534–0.557. Comparable incremental evidence for forecast-error magnitude is weak. Replication across independent portfolios is left as future research.

1. Motivation

Classical non-life reserving is usually presented through development triangles. Individual claims, however, contain information that triangles necessarily compress: reporting age, payment history, current status, and - importantly here - the case estimate held on an open claim. The question is not whether individual models are automatically superior. It is more precise: when aggregate and claim-level approaches diverge, can the divergence be traced to information that aggregation discards or uses only crudely?

This experiment was prompted by that practical distinction. The portfolio is synthetic, so the future is hidden from the models but known to the experimenter. That permits an unusually clean comparison: every method is evaluated at the same valuation cut against the same subsequently observed cash flows. The design is close in spirit to the individual paid-incurred reserving literature of Pigeon, Antonio and Denuit (2014), while the macro-versus-micro framing connects naturally to Charpentier and Pigeon (2016).

The central empirical result is not simply that a bottom-up model wins. Raw case reserves and incurred Chain Ladder are among the worst methods in this particular simulated provision regime. The useful information lies in the case estimate as a claim-level signal; it becomes valuable only after its systematic level distortion is modelled. In this study that provision process is generated by SPLICE; the result should therefore be read as conditional on the simulated case-reserving regime, not as a general indictment of incurred Chain Ladder.

2. A short reserving primer

2.1 Claim development and the valuation cut

A claim develops in calendar time. It occurs, is reported, may generate one or more payments, receives case-estimate revisions, and is eventually closed. At a valuation date T , only events at or before T are admissible predictors. Everything after T is the hidden future used for scoring.

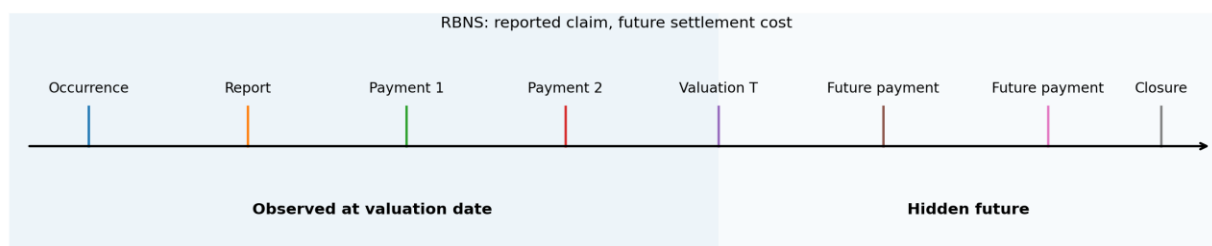


Figure 1 — Claim development around a valuation date. The valuation cut separates information available to the model from the future cash flows used only for validation.

2.2 RBNS, IBNR, paid and incurred

Reported But Not Settled (RBNS) refers here to future payments on claims that have already been reported by the valuation date. Incurred But Not Reported (IBNR) refers to future cost from claims that have occurred but have not yet been reported. The total reserve target is therefore decomposed as:

Total reserve = RBNS + IBNR. Throughout this note, 'reserve' means a best-estimate reserve: the conditional expected value of future claim payments given the information available at the valuation date. The target is therefore a mean monetary liability, not a quantile or capital measure.

Paid data record cash already disbursed. Incurred data combine paid amounts with an outstanding case estimate. In the SPLICE data used here, the outstanding case liability (OCL) obeys the accounting identity incurred = cumulative paid + OCL. OCL is therefore a case reserve or case estimate for the unpaid part of a reported claim; it is not the same object as IBNR.

2.3 Macro versus micro reserving

A macro method first aggregates claim development into a run-off triangle and then models the aggregate pattern. Chain Ladder is the canonical example. A micro or bottom-up method retains individual claim states, predicts future cost claim by claim, and sums those predictions. Aggregation can reduce noise and confer robustness, but it also discards detail. The empirical question is whether the discarded detail contains stable predictive information.

The research question: what predictive information is lost by aggregation?

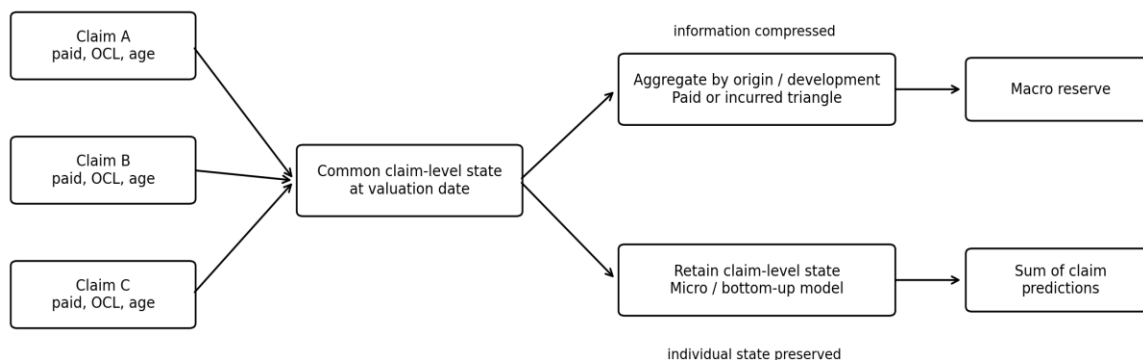


Figure 2 — Macro and micro reserving use the same underlying claims but preserve different information. A triangle deliberately compresses individual claim state.

3. Data and controlled experimental design

The study combines synthetic transaction histories with SPLICE incurred histories. The incurred file contains 31,250 history rows for 3,624 claims. Reconciliation is exact to numerical precision: the maximum absolute discrepancy in incurred - cumulative paid - OCL is approximately 10^{-8} , and claim-level paid histories reconcile with the transaction data.

Rolling pseudo-valuations are constructed so that models see only the historical state available at each cut. The final common evaluation window uses years 7, 8, 9 and 10. These are not independent portfolios: they are correlated valuation dates from one simulated book. This matters for interpretation and is discussed in the limitations.

3.1 As-of-T discipline and leakage control

At each pseudo-valuation date, an open claim is represented only by information observed by that date. Future payments, future case-estimate revisions and closure information are excluded from predictors. Model calibration - including the smearing factor described later - is re-estimated inside each historical training sample. The hidden future is used only after fitting, for evaluation.

4. Censoring: why the micro problem is harder than it looks

At an early valuation date, long-lived claims have not yet revealed their eventual settlement cost. Their outcomes are right-censored by the observation window. If a training set simply retains claims whose complete future is already observed, it preferentially selects short-lived claims. The resulting complete-case sample is not representative of the open claims that must be reserved.

This mechanism appeared clearly in the exploratory stages of the notebook. Claims excluded by the complete-outcome rule had substantially longer remaining lifetimes and larger remaining costs. At a six-year horizon, for example, excluded claims had roughly 2.25 times the mean remaining cost of eligible claims. This is selection induced by censoring, not random missingness.

Inverse Probability of Censoring Weighting (IPCW) attempts to correct this selection by weighting observed complete outcomes by the inverse probability that they would have been observed. Survival and payment-process variants were also explored. Those branches were methodologically useful but did not produce the final competitive estimator; they are retained in the computational notebook as negative results and diagnostics rather than presented as headline models.

5. Case estimates: informative signal, poor level

5.1 The year-10 state

At $T = 40$ quarters (ten years), 846 reported claims are open. Their OCL totals EUR 104.2 million, whereas their actual subsequent RBNS payments total EUR 174.0 million: raw OCL understates the aggregate future by 40.1%. Yet OCL is strongly associated with which claims will ultimately cost more. Pearson correlation with future RBNS is 0.604 and Spearman rank correlation is 0.599. Paid-to-date is far weaker: Pearson -0.029 and Spearman 0.157.

This distinction between ranking and level is central. A variable can order claims usefully while being badly calibrated as a monetary forecast.

5.2 Calibration by OCL decile

The decile analysis makes the distortion visible. Low-OCL claims subsequently cost many multiples of their case estimate, while the highest OCL decile is mildly over-reserved on average: its mean future/OCL ratio is about 0.95. The relationship is therefore strongly state-dependent, not a simple constant under-reserving factor at claim level. This shape helps explain why a learnable claim-level correction can be useful even when raw OCL is poor as a reserve amount.

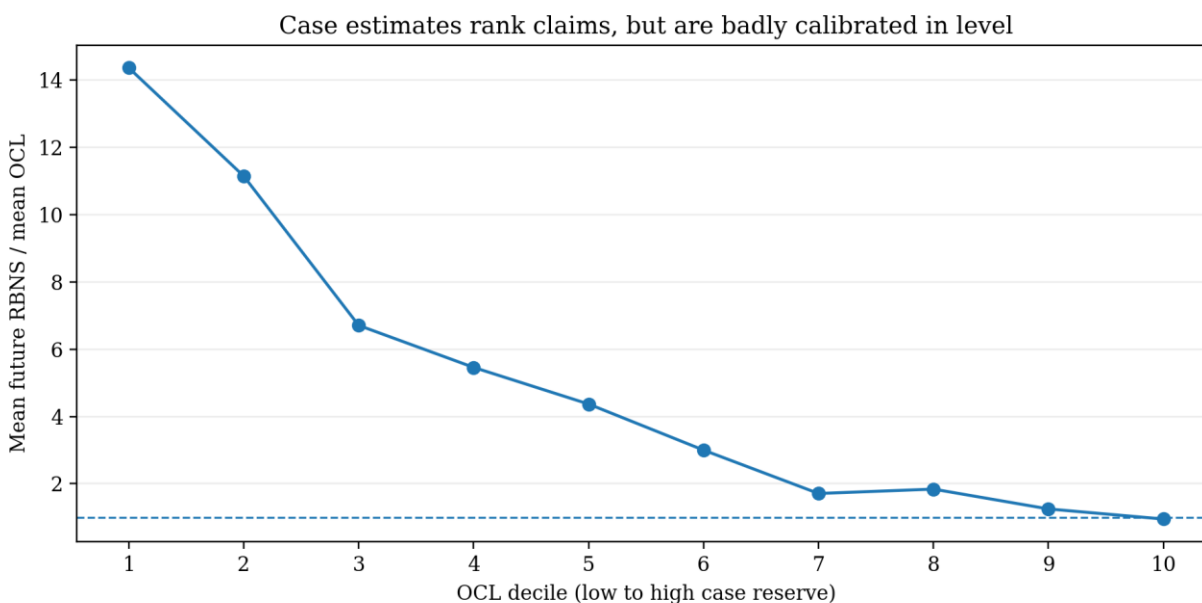


Figure 3 — Mean future RBNS divided by mean OCL by OCL decile at year 10. OCL contains strong ranking information but is systematically miscalibrated, especially among low-reserve claims.

6. The claim-level models

6.1 Paid-only baseline

The paid-only micro model uses claim information available at the valuation date but excludes OCL. On the common years 7-10 window it underestimates RBNS, with mean relative bias -19.3% and RBNS MAPE 19.3%. The exploratory recursive payment-process models were even less stable because long-horizon recursion, finite forecast caps and feature clipping can create extrapolation pathologies. They are not used to claim an informational advantage for case estimates.

6.2 Stable case-aware log model

Direct Gamma modelling of future claim cost produced severe extrapolation in one rolling window. A more stable specification therefore models $\log(1 + \text{future RBNS})$ as a function of claim state, including OCL. This substantially improves claim-level absolute error, but naive exponentiation creates a new problem: a regression fitted on a logarithmic scale does not directly estimate the arithmetic conditional mean required for reserving.

7. Retransformation and Duan smearing

7.1 Why exponentiating a log prediction is biased for the mean

Let Y denote future RBNS cost and X the observed claim state. Suppose the model is fitted as

$$\log(1 + Y) = m(X) + \varepsilon.$$

Exponentiating the fitted conditional location gives $\exp(m(X)) - 1$. But the reserve target is the conditional arithmetic mean. Because exponentiation is nonlinear,

$$E[Y | X] = \exp(m(X)) E[\exp(\varepsilon) | X] - 1,$$

not merely $\exp(m(X)) - 1$. Jensen's inequality explains the direction: for non-degenerate residual variation, $E[\exp(\varepsilon)]$ exceeds $\exp(E[\varepsilon])$. Even residuals centred at zero on the log scale therefore imply an upward retransformation correction on the original euro scale.

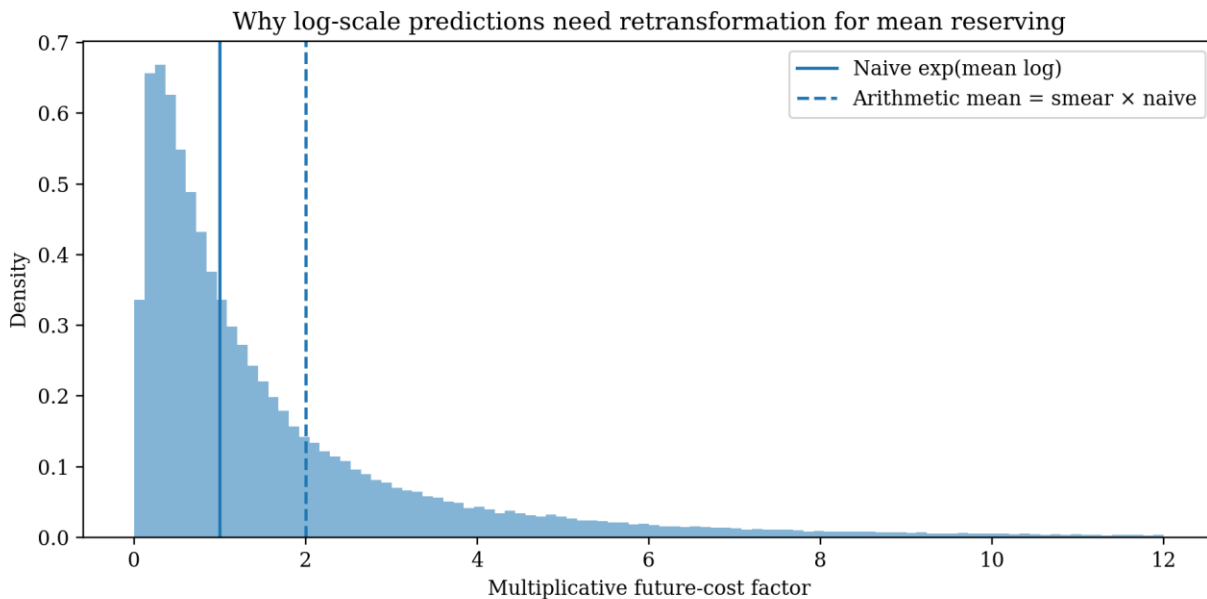


Figure 4 — Pedagogical illustration of retransformation. The centre of a log-scale model lies below the arithmetic mean on the original, right-skewed scale.

7.2 Duan's nonparametric smearing estimator

Duan (1983) proposed estimating the retransformation factor directly from fitted residuals rather than assuming a parametric residual distribution. For historical training observations, define residuals

$$\hat{\epsilon}_i = \log(1 + Y_i) - \hat{m}(X_i).$$

The global smearing factor is

$$\hat{S} = (1/n) \sum \exp(\hat{\epsilon}_i),$$

and the mean-scale prediction becomes

$$\hat{Y}_i(\text{smear}) = \hat{S} \exp(\hat{m}(X_i)) - 1.$$

In this experiment the rolling historical smearing factors are 1.997, 2.038, 2.032 and 2.022 for years 7-10. The factors themselves are strikingly stable across the four expanding training samples. That does not mean that the resulting reserve errors are stable: the case-aware RBNS errors are -12.6%, -5.2%, -0.7% and +18.2%, respectively. In particular, the year-10 model overshoots despite having the largest training sample. The factor is nevertheless not an ex-post multiplier chosen to repair the scored reserve; it is estimated separately from historical training residuals before the corresponding hidden future is revealed. The near-zero mean bias across the four dates must therefore be read together with this dispersion, not as evidence of uniformly exact calibration.

Year	Training states	Smearing factor	Case-model RBNS error
7	1,006	1.997	-12.6%
8	1,745	2.038	-5.2%
9	2,552	2.032	-0.7%
10	3,380	2.022	+18.2%

Reading Table 1: the smearing factor is stable; the forecast error is not. Across the four dates the case-model RBNS error has mean -0.1%, mean absolute error 9.2 percentage points, and ranges from -12.6% to +18.2%. The year-10 overshoot was therefore examined without refitting the model. The published year-10 fit was first reproduced exactly, after which its error was decomposed. The miss is diffuse rather than tail-driven: the 20 largest positive claim errors account for only 9.6% of total positive error. Miscalibration is concentrated mainly in OCL deciles 7-9, while lower OCL deciles remain underpredicted. A further log-residual diagnostic separates two mechanisms. Mean log residuals vary materially by OCL decile (from -0.273 to +0.623), showing conditional-mean misspecification. Residual dispersion also declines sharply with OCL: the standard deviation of log residuals falls from 1.831 in the lowest OCL decile to about 0.50 in the highest deciles. Thus both the fitted mean and the conditional uncertainty structure vary with claim state. The ex-post quantity $\text{mean exp}(\log \text{residual})$ is consequently treated only as a total multiplicative correction, because it combines location error and dispersion rather than identifying a pure retransformation factor.

7.3 Why smearing can improve the portfolio mean while worsening individual MAE

The naive log model was attractive for typical individual claims but severely understated the aggregate. Smearing changes the estimand: it targets the arithmetic conditional mean rather than the log-scale centre. For a best-estimate reserve, the portfolio quantity of interest is the sum of conditional means. A method can therefore have a somewhat larger claim-level MAE yet a much better aggregate reserve.

8. Common-window horse race

All headline methods are compared on the same four pseudo-valuations, years 7-10, against the same hidden total reserve. RBNS estimators are placed on the total-reserve target by adding exactly the same causal IBNR estimate. This is useful operational presentation, but it does not constitute independent evidence: among the RBNS-based methods, Table 2 is essentially the RBNS comparison of Table 3 plus a shared IBNR component. The cleanest information comparison therefore lives in the RBNS-only decomposition.

Method	Mean bias	MAPE	RMSE (EUR m)	Worst error
Case-aware global smear + IBNR	-1.2%	9.4%	21.5	16.4%
Paid Chain Ladder	+3.6%	13.2%	30.8	22.6%
Paid-only micro + IBNR	-18.3%	18.3%	39.6	23.3%
Raw OCL + IBNR	-36.0%	36.0%	73.2	39.6%
Incurred Chain Ladder	-48.2%	48.2%	98.2	53.5%

Table 2 is an operational total-reserve presentation. For Raw OCL, paid-only micro and case-aware micro, the same causal IBNR estimate is added to the RBNS estimate. Their relative ranking therefore derives primarily from the RBNS component shown separately in Table 3.

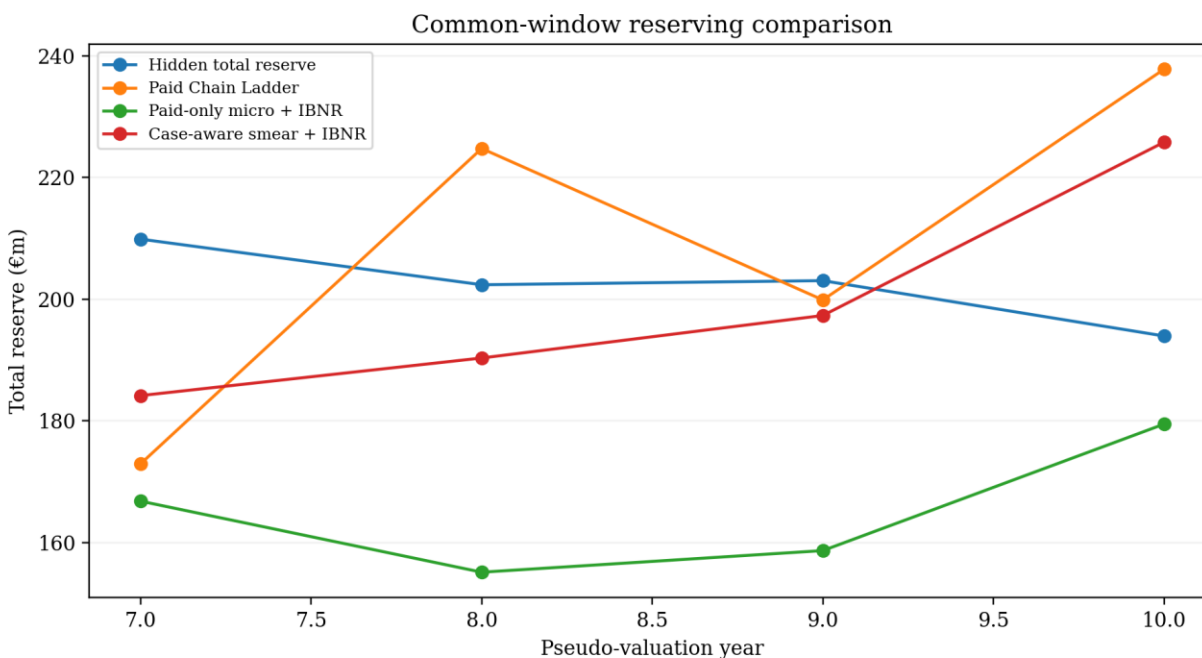


Figure 5 — Common-window total-reserve comparison. The case-aware model is closest overall, while paid Chain Ladder remains a strong benchmark.

8.1 RBNS-only decomposition

Removing IBNR makes the information comparison sharper. On RBNS alone, the case-aware smearing model has mean relative bias -0.1%, MAPE 9.2% and RMSE EUR 20.4 million. The paid-only micro model has bias -19.3%, MAPE 19.3% and RMSE EUR 37.1 million. Raw OCL has mean bias -39.0% and MAPE 39.0%. The case-aware model's near-zero mean bias should not be mistaken for uniformly small bias: its four annual errors range from -12.6% to +18.2%, so cancellation across dates matters. Descriptively, on this one portfolio, adding and correctly retranslating the case-estimate information roughly halves RBNS MAPE relative to the paid-only claim-level model.

RBNS method	Mean bias	MAPE	RMSE (EUR m)	Worst error
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Case-aware global smear	-0.1%	9.2%	20.4	18.2%
Paid-only micro	-19.3%	19.3%	37.1	24.6%
Raw OCL	-39.0%	39.0%	71.1	42.8%

9. Interpretation

The ordering of the methods is more informative than a simple claim that bottom-up reserving beats Chain Ladder. In this simulated provision regime, raw OCL is badly low and incurred Chain Ladder is worse still. Yet OCL is a much stronger claim-level ranking signal than paid-to-date. Once the case estimate is retained at claim level and the log-scale prediction is retransformed toward the mean monetary target, the information becomes useful enough, on this portfolio and common window, to outperform paid Chain Ladder.

A positive finding is therefore provision-regime dependence: incurred Chain Ladder inherits the quality and dynamics of the case-reserving process. In this SPLICE regime, systematic under-reserving in case estimates propagates into the incurred triangle; aggregation does not repair it. This should not be generalized into 'incurred Chain Ladder is bad'. Rather, it identifies a condition under which incurred triangles can fail and a reason to diagnose the underlying case-reserving process. Conversely, an individual model can exploit a biased case estimate if the relationship between that estimate, claim state and future cost is learnable from history.

This is closely related to Pigeon, Antonio and Denuit (2014), who explicitly extend individual payments-only reserving to paid-incurred information. The present work is not a replication of their iPIC model; rather, it provides a controlled computational experiment around the same substantive idea: incurred information at individual-claim level can improve prediction when it is modelled as information rather than accepted uncritically as truth.

9.1 Incremental information in the case estimate

The raw association between OCL and future RBNS does not by itself measure what the case estimate adds, because OCL is correlated with paid-to-date and claim maturity. A final added-variable diagnostic therefore projects $\log(1 + \text{OCL})$ on the paid-state variables used here - paid-to-date, claim age and report age - and defines $\text{OCL}_{\perp}$ as the residual case-estimate signal. The paid-only prediction error is centred within each valuation date so that the known aggregate under-reserving bias of that benchmark does not masquerade as incremental mean information.

The incremental level signal is substantial and stable. Pooled over years 7-10, $\text{OCL}_{\perp}$ has partial Pearson correlation 0.384 and partial Spearman correlation 0.544 with the date-debiased signed paid-model residual; a one-variable linear added-variable projection has descriptive partial R^2 0.147. The year-specific Spearman correlations are 0.557, 0.546, 0.534 and 0.550. The decile pattern is approximately monotone: mean signed error moves from EUR -184 thousand in the lowest $\text{OCL}_{\perp}$ decile to EUR +326 thousand in the highest. Thus, in this experiment, the individual case estimate contains material information about future RBNS level that is not explained by the paid-state variables used in the diagnostic.

The corresponding incremental uncertainty result is weak. For log absolute date-debiased error, the pooled partial Pearson correlation is -0.031, partial Spearman correlation -0.147 and linear partial R^2 0.001; year-specific partial R^2 values are essentially zero. Raw OCL remains associated with heteroscedastic residuals, but the present experiment does not support the stronger claim that the component of OCL incremental to the paid-state variables carries substantial additional information about forecast-error magnitude.

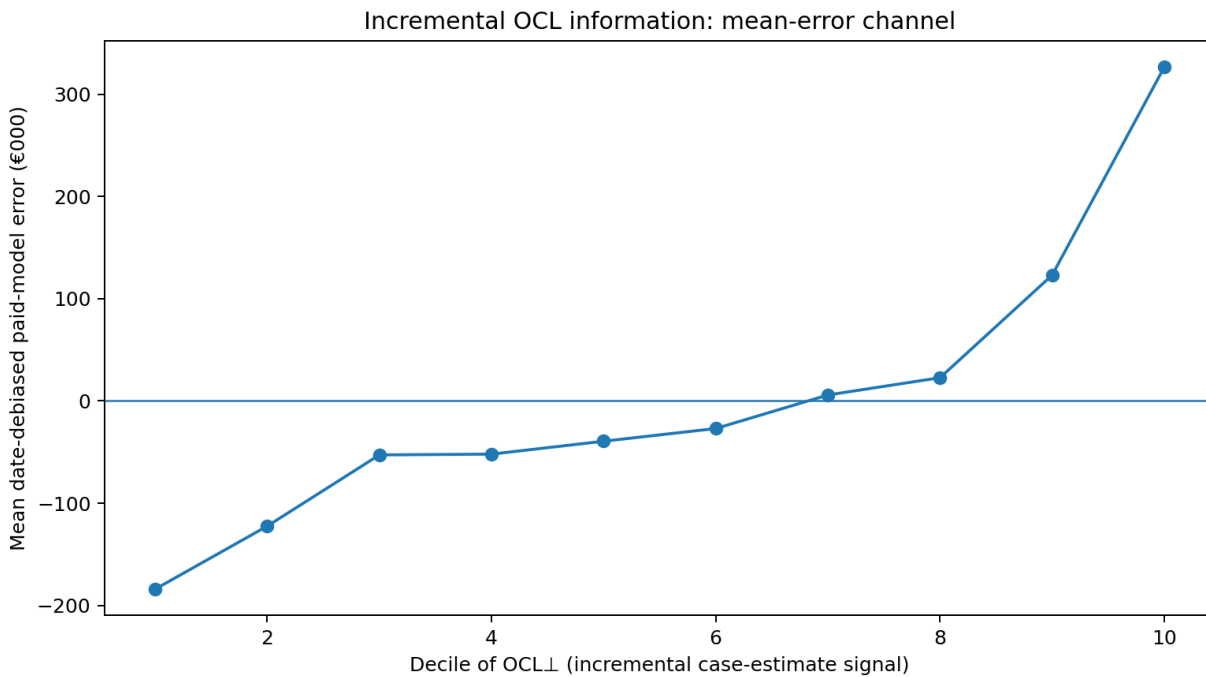


Figure 6 — Incremental information in the case estimate. Claims are grouped by decile of OCL⊥, the component of log OCL not explained by paid-to-date, claim age and report age. The vertical axis is the mean paid-only prediction error after removing valuation-date mean bias. The approximately monotone relationship shows material incremental information about future RBNS level.

10. Limitations and next validation

- One simulated portfolio. The four headline valuation dates are correlated observations from the same book, not four independent experiments.
- Provision-regime dependence. The poor incurred Chain-Ladder result is specific to the materially under-reserved SPLICE case-estimate process used here. The experiment demonstrates transmission of provision quality into incurred-triangle performance, not general inferiority of incurred Chain Ladder.
- Paid-process caveat. Recursive payment-process branches use a finite future-quarter cap and feature clipping to training support; these safeguards can induce downward bias in long-lived claims. Those branches are diagnostic, not headline evidence.
- **Conditional calibration. The year-10 diagnostic shows both log-scale mean misspecification and heteroscedastic residuals across raw OCL strata. The subsequent incremental-information diagnostic qualifies the interpretation: after conditioning on the paid-state variables used here, OCL⊥ retains strong information about signed mean error but little linear explanatory power for error magnitude. Accordingly, the present note does not claim that case estimates add substantial incremental uncertainty information beyond those paid-state variables.**

Future research. The principal complementary study is replication across independently generated SynthETIC/SPLICE portfolios with the present modelling and diagnostic procedure frozen in advance. Replication should test the stability of the incremental OCL level signal, the case-aware-versus-paid Chain Ladder performance gap, and the raw heteroscedastic pattern. Distributional or heteroscedastic claim models - for example a GAMLSS-style formulation with both location and scale depending on

claim state - remain a principled modelling direction, but the current incremental diagnostic does not establish that OCL adds substantial uncertainty information beyond paid-state variables. Such models should first be tested on fresh portfolios, not fitted post-hoc to this already-inspected book. A stronger second stage would vary the case-reserving regime, use a second simulator, or use real claims data with trustworthy historical case-estimate snapshots.

11. Conclusions

Five conclusions are worth retaining.

- Case estimates can be highly informative without being well calibrated. At year 10, OCL has Spearman correlation about 0.60 with future RBNS while understating aggregate RBNS by about 40%.
- Naive use of incurred information is not enough. Raw OCL and incurred Chain Ladder perform poorly in this simulated under-reserving regime.
- The log transformation solves a stability problem but creates a retransformation problem. For reserving, the target is an arithmetic conditional mean, not the centre of the log-scale response.
- Retransformation is consequential rather than cosmetic, but it is not the whole story. Historical Duan factors near 2.0 materially affect the mean-scale reserve, while the year-10 residual diagnostic shows both conditional-mean error and heteroscedasticity across raw OCL strata.
- The case estimate adds measurable information about RBNS level beyond the paid-state variables used here. After partialling log OCL on paid-to-date, claim age and report age, OCL $\perp$ has pooled partial Spearman correlation 0.544 with date-debiased paid-model error, stable at 0.534-0.557 across years. Comparable incremental evidence for error magnitude is weak (partial R^2 0.001).
- On this one synthetic portfolio and the common years 7-10 window, the case-aware model records 9.4% total-reserve MAPE versus 13.2% for paid Chain Ladder. On RBNS alone, its MAPE is 9.2% versus 19.3% for the paid-only micro model. The total-reserve comparison reuses the same causal IBNR component for RBNS-based methods and is therefore not a second independent confirmation. These descriptive results motivate replication across independent portfolios; they are not a theorem of micro-model superiority.

12. Glossary and notation

Term	Meaning
Case estimate / case reserve	An adjuster or insurer estimate, at a given valuation date, of the unpaid amount expected on a reported claim.
Censoring	Incomplete observation of an outcome because the observation window ends before the claim has fully developed.
Chain Ladder (CL)	A classical macro reserving method that projects aggregate run-off triangles through development factors.
Complete-case analysis	Analysis restricted to observations whose outcomes are fully observed. Under censoring this can create selection bias.
IBNR	Incurred But Not Reported: reserve for claims that have occurred but are not yet reported at the valuation date.
Incurred	For this study, cumulative paid plus outstanding case liability (OCL).
IPCW	Inverse Probability of Censoring Weighting: weights observed outcomes by the

	inverse probability that they are observable, to correct censoring selection under suitable assumptions.
MAE	Mean Absolute Error: average absolute prediction error in monetary units.
MAPE	Mean Absolute Percentage Error: average of absolute relative errors. Here it summarizes reserve error across pseudo-valuation dates.
Macro / top-down reserving	Reserving from aggregated development data, typically triangles.
Micro / bottom-up reserving	Reserving from individual claim histories or states, with predictions summed to portfolio level.
OCL	Outstanding Case Liability: the SPLICE field representing the outstanding case estimate; incurred = cumulative paid + OCL.
Out-of-time validation	Fitting only on information available before a valuation cut and scoring on subsequently revealed outcomes.
Paid	Cash actually disbursed on claims by the valuation date.
Pearson correlation	Linear correlation between two numerical variables.
Pseudo-valuation date	A historical cut imposed on complete simulated data to reproduce what would have been known at that time.
RBNS	Reported But Not Settled: future settlement cost of claims already reported at the valuation date.
RMSE	Root Mean Squared Error: square root of mean squared monetary error; it penalizes large misses more heavily than MAE.
Smearing estimator	Duan's nonparametric correction for retranslating predictions from a transformed regression scale to an expected response on the original scale.
Spearman correlation	Rank correlation: measures monotone ordering rather than strictly linear association.
Best-estimate reserve	Conditional expected value of future claim payments given information available at the valuation date; the mean liability targeted in this note.
Estimand	The statistical quantity a model is intended to estimate. Here the central estimand is the conditional arithmetic mean of future claim payments.
Jensen's inequality	For a convex function such as $\exp$, $E[\exp(Z)]$ is at least $\exp(E[Z])$; this is why naive exponentiation of a log-scale prediction generally misses the mean on the original scale.

13. References

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Appendix A. Computational notebook architecture

The companion notebook is `reserving_triangles_v2.8.0_incremental_ocl_information`. It is a research log as well as an executable experiment. A switchboard near the beginning controls expensive branches. The final case-estimate study, OCL recalibration, smearing study, common-window horse race, year-10 error anatomy, log-residual mechanism diagnostic and incremental OCL information diagnostic are retained; unsuccessful recursive payment-process branches remain available for auditability but are not headline evidence.

The notebook develops the study in stages: synthetic paid triangles and Chain Ladder; claim-level paid models; rolling out-of-time backtests; censoring diagnostics and IPCW; survival and payment-process experiments; SPLICE case-estimate reconciliation; raw OCL diagnostics; stable case-aware log modelling; Duan smearing; the common-window comparison; and a fixed-model year-10 error decomposition. The paper intentionally compresses this research history into a teachable statistical argument rather than reproducing every exploratory cell.

Reproducibility note. The headline comparison uses years 7-10 only because these dates are shared by all final methods. At each date the hidden future is used only for scoring. The smearing factor is estimated from historical training residuals inside the rolling fit. The year-10 diagnostic first reproduces the published fit exactly (3,380 training states, smearing factor 2.022, predicted RBNS EUR 205.6 million) and only then decomposes its error. A subsequent log-residual analysis distinguishes conditional-mean error from heteroscedasticity; neither diagnostic is used to refit year 10. Section 9.1/§30 is diagnostic only: OCL is residualised on paid-state variables, and its association with the frozen paid-model residual is measured without feeding OCL back into a new reserving model. The signed residual is centred by valuation date before the incremental level test.

Data provenance. Synthetic claims: SynthETIC; incurred/case-estimate process: SPLICE.